**Proposition.** (Weak Runge's Theorem — case of $E = \mathbb{C}_\infty - K$) ← largest possible

Let $G \subset \mathbb{C}$ be an open set, and let $K \subset G$ be a compact subset.

Then for any holomorphic function $f: G \to \mathbb{C}$ on $G$ and any $\epsilon > 0$,

there exists a rational function $R(z)$ whose poles lie in $\mathbb{C}_\infty - K$ — in fact all in $G \setminus K$ ($\infty$ not needed)

such that $|f(z) - R(z)| < \epsilon$ for all $z \in K$ uniformly on $K$

*Proof.* Recall:

**Proposition.** (Separation Lemma)

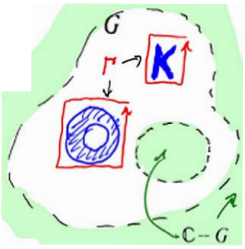
*Let $G \subset \mathbb{C}$ be an open set, and let $K \subset G$ be a compact subset.*

Then there exists a *1-cycle* $\Gamma$ *in* $G - K \subset G$ such that

$$n(\Gamma; w) = 0 \quad \textit{for all} \quad w \in \mathbb{C} - G$$

$$\textit{and} \quad n(\Gamma; z) = 1 \quad \textit{for all} \quad z \in K$$

Moreover, $\Gamma$ consists of *a finite number of closed polygons.*




(Enhanced Separation Lemma)

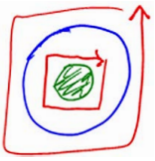
compare

*Chapter VIII*

**1.1 Proposition.** *Let $K$ be a compact subset of the region $G$; then there are straight line segments $\gamma_1, \ldots, \gamma_n$ in $G - K$ such that for every function $f$ in $H(G)$,*

$$f(z) = \sum_{k=1}^{n} \frac{1}{2\pi i} \int_{\gamma_k} \frac{f(w)}{w - z}\, dw$$

*for all $z$ in $K$. The line segments form a finite number of closed polygons.*



**1.7 Runge's Theorem.** for compact subsets of open sets

Let $G \subset \mathbb{C}$ be an open set, and let $K \subset G$ be a compact subset.

Let $E \subset \mathbb{C}_\infty - K$ be a subset that meets each component of $\mathbb{C}_\infty - K$. (open in $\mathbb{C}_\infty$)

Then for any holomorphic function $f: G \to \mathbb{C}$ on $G$ and any $\epsilon > 0$,

there exists a rational function $R(z)$ whose poles lie only in $E$

such that $|f(z) - R(z)| < \epsilon$ for all $z \in K$

($R$ approximates $f$; uniformly on $K$)

*Proof.* First, recall some standard results:

- **Lemma 0** If $\{f_n\} \to f$, $\{g_n\} \to g$ uniformly on $K$, then $\{f_n + g_n\} \to f + g$ uniformly on $K$

- **Lemma 1** If there exists a sequence $\{f_n\} \to f$ uniformly on $K$, and for each $n$, there exists a sequence $\{g_j^{(n)}\} \to f_n$ uniformly on $K$, then there exists a subsequence of the double sequence $\{g_j^{(n)}\}$ which converges to $f$ uniformly on $K$

  ($\forall j \geq j(n)$, $\| g_j^{(n)} - f_n \|_K < \frac{1}{n}$; then $\left( g_{j(n)}^{(n)} \right)_n \to f$ uniformly on $K$)

- **Lemma** (partial fractions expansion)

  Let $R(z) = \dfrac{p(z)}{q(z)}$ be a rational function where $p(z), q(z)$ are polynomials without common factors.

  Suppose the unique factorization of $q(z)$ in $\mathbb{C}[z]$ is

  $$q(z) = c(z-a_1)^{k_1} \cdots (z-a_m)^{k_m} \quad \text{with } c \in \mathbb{C}^\times,\ k_j \in \mathbb{N}_{>0},\ a_1, \ldots, a_m \in \mathbb{C} \text{ pairwise distinct}$$

  Then there exist unique polynomials $P \in \mathbb{C}[z]$ and $g_1, \ldots, g_m \in \mathbb{C}[z]$ with $\deg g_j < k_j$

  such that $$R(z) = P(z) + \sum_{j=1}^{m} \frac{g_j(z)}{(z-a_j)^{k_j}} \quad \text{in } \mathbb{C}(z)$$

  ($P$: rat. fn. whose pole is at $\infty$)

- We have already shown

**Proposition.** (Weak Runge's Theorem — case of $E = \mathbb{C}_\infty - K$) (largest possible)

Let $G \subset \mathbb{C}$ be an open set, and let $K \subset G$ be a compact subset.

Then for any holomorphic function $f: G \to \mathbb{C}$ on $G$ and any $\epsilon > 0$,

there exists a rational function $R(z)$ whose poles lie in $\mathbb{C}_\infty - K$ (not $\infty$)

such that $|f(z) - R(z)| < \epsilon$ for all $z \in K$

(uniformly on $K$)

$c z^k$ $\quad \frac{c}{(z-a)^k}$

- By **Weak Runge's Theorem** and **Lemma 1**,

  we may assume the holomorphic function $f: G \to \mathbb{C}$

  is a rational function $f \in \mathbb{C}(z)$ whose poles lie in $\mathbb{C}_\infty - K$

- By partial fractions expansion and **Lemma 0**, ($\infty$ not appearing)

  we may assume further that $f \in \mathbb{C}(z)$ has only one pole $z_0$, and $z_0 \in \mathbb{C}_\infty - K$

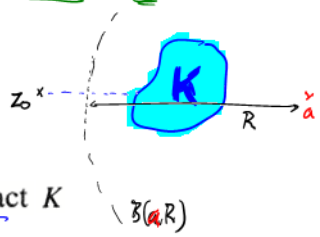
  i.e. $f(z) = \dfrac{c}{(z-z_0)^n}$ with $c \in \mathbb{C}^\times$, $z_0 \in \mathbb{C}_\infty - K$ (not $\infty$)

- By hypothesis, there exists a point $w \in E$ in the component $U$ of $\mathbb{C}_\infty - K$ containing $z_0$

**We want to** approximate $f$ uniformly on $K$ using rational functions with only one pole at $w$

- If $w = \infty$, then choose $a \in \mathbb{C}$ and $R > 0$ such that

  $K \subset B(a; R)$ and $z_0 \notin B(a; R)$




Then the power series expansion of $f$ centered at $a$

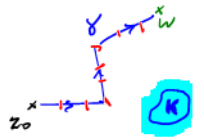
converges normally on compact $K$

$\Rightarrow$ the partial sums (of pwr series) are **polynomials** converging to $f$ uniformly on $K$

(rat. fns with pole only at $w=\infty$)

- Suppose $w \in \mathbb{C}$. (so $w \neq \infty$) Since $U \subset \mathbb{C}_\infty$ (open) is locally path connected,

  we can join $z_0$ to $w$ ($*\infty*$) by a polygonal path $\gamma$ in $U$




Let $\delta := d(K, \gamma) > 0$

Write $\gamma = \gamma_0 + \cdots + \gamma_n$ as a finite sum of straight line segments $\gamma_j = [z_j, z_{j+1}]$ in $U$

each of length $< \delta$ $\quad (w = z_{n+1})$

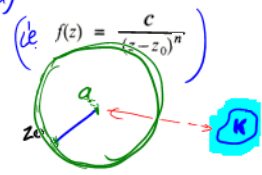
Reduced to show:

- **Lemma** ("pole pushing")

Suppose $f \in \mathbb{C}(z)$ (rat. fn) has only one pole $z_0$, and $z_0 \in \mathbb{C}_\infty - K$ ($\infty$ not allowed)

Then for any $a \in \mathbb{C}_\infty - K$ such that $[a, z_0] \subset \mathbb{C}_\infty - K$

and $|a - z_0| < d(K, [a, z_0])$




there exists a sequence of rational functions with only one pole at $a \in \mathbb{C}_\infty - K$

converging to $f$ uniformly on $K$

*Proof.* We have $K \subset \mathrm{ann}(a; |a - z_0|, \infty)$ open annulus

The Laurent series expansion of $f$ centered at $a$

converges normally on compact $K$

Its holomorphic part vanishes at $\infty$ (same for $f$)

and is an entire function

hence is $= 0$ by Liouville's Theorem

$\Rightarrow$ the partial sums are rational functions with only one pole at $a$

(of Laurent series $c_k/(z-a)^k$) converging to $f$ uniformly on $K$

$\#$

**1.14 Corollary.** (Runge's Approximation on open sets)

Let $G \subset \mathbb{C}$ be an open set

Let $E \subset \mathbb{C}_\infty - G$ be a subset that meets each component of $\mathbb{C}_\infty - G$.

Then for any **holomorphic function** $f: G \to \mathbb{C}$ on $G$

there exists a sequence of rational functions whose poles lie only in $E$

converging to $f$ locally uniformly on $G$

compare: **1.14 Corollary.** *Let $G$ be an open subset of the plane and let $E$ be a subset of $\mathbb{C}_\infty - G$ such that $E$ meets every component of $\mathbb{C}_\infty - G$. Let $R(G,E)$ be the set of rational functions with poles in $E$ and consider $R(G,E)$ as a subspace of $H(G)$. If $f \in H(G)$ then there is a sequence $\{R_n\}$ in $R(G,E)$ such that $f = \lim R_n$. That is, $R(G,E)$ is dense in $H(G)$.*

**1.15 Corollary.** Let $G \subset \mathbb{C}$ be an open set such that $\mathbb{C}_\infty - G$ is connected (opt)

Then for any **holomorphic function** $f: G \to \mathbb{C}$ on $G$

there exists a sequence of **polynomials** converging to $f$

locally uniformly on $G$

(Take $E = \{\infty\}$)

compare **1.15 Corollary.** *If $G$ is an open subset of $\mathbb{C}$ such that $\mathbb{C}_\infty - G$ is connected then for each analytic function $f$ on $G$ there is a sequence of polynomials $\{p_n\}$ such that $f = \lim p_n$ in $H(G)$.*

compare

**2.2 Theorem.** *Let $G$ be an open connected subset of $\mathbb{C}$. Then the following are equivalent:*

(a) *$G$ is simply connected;*
(b) *$n(\gamma; a) = 0$ for every closed rectifiable curve $\gamma$ in $G$ and every point $a$ in $\mathbb{C} - G$;*
(c) *$\mathbb{C}_\infty - G$ is connected;*
(d) *For any $f$ in $H(G)$ there is a sequence of polynomials that converges to $f$ in $H(G)$;* ← Runge's thm
(e) *For any $f$ in $H(G)$ and any closed rectifiable curve $\gamma$ in $G$, $\int_\gamma f = 0$;*
(f) *Every function $f$ in $H(G)$ has a primitive;*
(g) *For any $f$ in $H(G)$ such that $f(z) \neq 0$ for all $z$ in $G$ there is a function $g$ in $H(G)$ such that $f(z) = \exp g(z)$;*
(h) *For any $f$ in $H(G)$ such that $f(z) \neq 0$ for all $z$ in $G$ there is a function $g$ in $H(G)$ such that $f(z) = [g(z)]^2$;*
(i) *$G$ is homeomorphic to the unit disk;*
(j) *If $u: G \to R$ is harmonic then there is a harmonic function $v: G \to R$ such that $f = u + iv$ is analytic on $G$.* ← Harmonic Fns

↳ IR-var mthds in $\mathbb{C}$-analysis

**2.2 Theorem.** ( Characterizations of simply connected open subsets of $\mathbb{C}$ )

*Let $G \subset \mathbb{C}$ be a connected open set.* The following are equivalent :

**(i)** $G$ is homeomorphic to $\mathbb{D}$

⇓

**(a)** $G$ is *simply connected*

topological prop of $G$

embedding prop $G$ into $\mathbb{C}$

**(b)** for any closed piecewise ~~smooth~~ $C^1$ curve $\gamma$ in $G$ and for all $w$ in $\mathbb{C} - G$, one has $n(\gamma; w) = 0$

(c) ⇒ (d) Runge's Approximation on open sets

**(c)** $\mathbb{C}_\infty - G$ is connected ⇒ Runge Approx. ⇒ **(d)** for any **holomorphic** function $f$ on $G$ there exists a sequence of **polynomials** converging to $f$ locally uniformly on $G$

extended complement

$\mathbb{C}$-analytic prop of $G$

**(e)** for any **holomorphic** function $f$ on $G$ and for any closed piecewise ~~smooth~~ $C^1$ curve $\gamma$ in $G$ one has $\int_\gamma f\,dz = 0$

(d) ⇒ (e) easy